Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-I Vascular Conditions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-09-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild the three patterns of arteriosclerosis

A competent response contains

· Arteriosclerosis means thickening and loss of elasticity of arterial walls and is the umbrella term for three patterns.

· Atherosclerosis: intimal disease of elastic and large to medium muscular arteries, with lipid, inflammatory cells, and fibrous tissue forming plaques that bulge into the lumen and narrow it.

· Hyaline arteriolosclerosis: small arteries and arterioles; plasma proteins leak across injured endothelium and smooth muscle cells lay down excess matrix, giving a homogeneous pink, glassy wall with a narrowed lumen.

· Hyaline form accompanies aging, long-standing benign hypertension, and diabetes; in the kidney it causes nephrosclerosis with a finely granular cortical surface, and in diabetes it affects both afferent and efferent arterioles.

· Hyperplastic arteriolosclerosis: severe or malignant hypertension; smooth muscle cells and reduplicated basement membrane form concentric onion-skin layers, often with fibrinoid necrosis, and the markedly narrowed lumen produces renal ischemia with pinpoint cortical hemorrhages.

· Monckeberg medial calcific sclerosis: calcification of the media and internal elastic lamina of medium muscular arteries, typically femoral, tibial, radial, and ulnar, in people over about 50.

· In Monckeberg sclerosis the intima and lumen are spared, so flow is not reduced; the vessel is palpably hard and appears as a pipestem on radiographs.

· Atherosclerosis is only one of the three patterns, so arteriosclerosis is the broader term.

Distinctions that matter

· Layer and vessel size separate the three: intima of large arteries, wall of arterioles, media of medium limb arteries.

· Only atherosclerosis and arteriolosclerosis reduce the lumen; Monckeberg sclerosis stiffens the wall without obstructing flow.

Common misconceptions

· Using arteriosclerosis and atherosclerosis as synonyms, which loses the two non-plaque patterns.

· Assuming calcification of an artery must mean reduced flow. Medial calcification leaves the lumen open and only stiffens the vessel.

Where it is taught: Section C01, all three paragraphs and the patterns table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C01-BD2

Brain dump 2 · integration and application: Two patients with hardened arteries

A competent response contains

· First patient: Monckeberg medial calcific sclerosis, with calcium in the media and internal elastic lamina; the intima and lumen are spared so flow is not reduced.

· A stiff calcified artery resists cuff compression, so the cuff pressure needed to stop flow exceeds the true intraluminal pressure and the ankle reading is falsely high.

· An index above 1.40 therefore means noncompressible arteries and an unreliable result, common in diabetes and chronic kidney disease, and it does not exclude coexisting atherosclerotic disease.

· Second patient: hyperplastic arteriolosclerosis of severe or malignant hypertension.

· The onion-skin layers are concentric smooth muscle cells and reduplicated basement membrane, often with fibrinoid necrosis of the wall.

· The markedly narrowed lumen reduces renal perfusion, producing ischemia with pinpoint cortical hemorrhages.

· A finely granular cortical surface with glassy pink arterioles is hyaline arteriolosclerosis with benign nephrosclerosis, from aging, long-standing benign hypertension, or diabetes.

Distinctions that matter

· A high index and a low index both come from a stenosis question, but a high index reports wall stiffness while a low index reports lost pressure across a narrowing.

· Hyaline and hyperplastic arteriolosclerosis affect the same vessels; the material in the wall and the severity of the hypertension separate them.

Common misconceptions

· Reading an index of 1.52 as excellent perfusion. Any value above 1.40 is uninterpretable.

· Expecting Monckeberg sclerosis to cause claudication. It does not narrow the lumen, so symptoms in such a patient come from atherosclerosis alongside it.

Where it is taught: Section C01, third paragraph and the patterns table, with the index bands in Section C06 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C01-CP01

Comparison table: The patterns of arterial hardening side by side

A competent response contains

· Vessels: atherosclerosis in the aorta and large to medium muscular arteries; both arteriolosclerosis forms in arterioles and small arteries; Monckeberg sclerosis in medium muscular arteries of the limbs.

· Layer: atherosclerosis is intimal; arteriolosclerosis involves the arteriolar wall with subendothelial protein and matrix or hyperplastic smooth muscle; Monckeberg sclerosis is medial with the internal elastic lamina.

· Lesion material: lipid core with a fibrous cap of smooth muscle and collagen; leaked plasma protein plus excess matrix giving a glassy pink wall; concentric smooth muscle and reduplicated basement membrane with fibrinoid necrosis; calcium.

· Lumen and flow: narrowed and liable to thrombose; narrowed; markedly narrowed; not narrowed and flow preserved.

· Setting: smoking, high LDL, hypertension, diabetes, and age; aging, benign hypertension, and diabetes; severe or malignant hypertension; age over about 50 with diabetes or chronic kidney disease.

· Ankle pressure: atherosclerotic stenosis lowers it and lowers the index; arteriolosclerosis of small vessels does not change a limb pressure measurement; Monckeberg calcification makes the artery noncompressible so the reading is falsely high and the index can exceed 1.40.

Distinctions that matter

· The first three patterns narrow the lumen while the fourth stiffens it, so only the fourth produces a falsely high ankle pressure.

· Hyaline disease tracks long-standing mild to moderate hypertension and diabetes; hyperplastic disease tracks a severe pressure rise.

Common misconceptions

· Putting Monckeberg sclerosis in the intima because it calcifies. The calcium sits in the media and internal elastic lamina.

· Treating fibrinoid necrosis as a feature of all arteriolosclerosis. It belongs to the hyperplastic form.

Where it is taught: Section C01, patterns of arteriosclerosis table, with the index bands in Section C06 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-BD1

Brain dump 1 · foundational retrieval: Build an atheroma from the first injury

A competent response contains

· Order: endothelial dysfunction at branch points and curvatures; LDL enters the intima and is oxidized; monocytes adhere, migrate in, and become macrophages; scavenger receptor uptake of oxidized LDL makes foam cells; fatty streak forms as a flat intimal collection of foam cells; PDGF and other growth factors recruit smooth muscle from the media; a fibrous cap of collagen covers a necrotic lipid core; rupture, erosion, hemorrhage, or calcification gives a complicated plaque.

· Macrophages use scavenger receptors, which are not down-regulated by cell cholesterol, so uptake of oxidized LDL continues until the cell becomes a foam cell.

· Endothelial dysfunction is promoted by hypertension, smoking, hyperlipidemia, hyperglycemia, and inflammation, and the dysfunctional endothelium becomes more permeable and expresses VCAM-1.

· A mature atheroma has a fibrous cap of smooth muscle cells and collagen over a core of lipid, necrotic debris, foam cells, and cholesterol crystals.

· Smooth muscle cells recruited from the media build the cap.

· Distribution in descending order: abdominal aorta more than thoracic, coronary arteries, popliteal arteries, internal carotid arteries, circle of Willis.

· Modifiable risks: smoking, hypertension, raised LDL, diabetes. Fixed risks: age, male sex, family history.

· Plaques are patchy rather than uniform along a vessel.

Distinctions that matter

· Scavenger receptor uptake is unregulated, unlike LDL receptor uptake, which is why macrophages overload.

· The fatty streak is a flat foam cell collection and not obstructive; the mature plaque bulges into the lumen.

Common misconceptions

· Describing atherosclerosis as simple lipid deposition. It is a chronic inflammatory response of the wall to endothelial injury.

· Expecting plaque where flow is fastest. It forms where flow is disturbed rather than laminar, at branch points and curvatures.

Where it is taught: Section C02, opening paragraph and the atherogenesis flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-BD2

Brain dump 2 · integration and application: Why a small plaque can kill and a big one can ache

A competent response contains

· Patient A: a lipid-rich plaque with a thin, inflamed cap ruptured, exposing thrombogenic contents, and the overlying thrombus occluded the artery, so the event depended on stability rather than size.

· Patient B: a stable plaque narrowed the lumen gradually, and once stenosis reaches about 70% flow cannot rise to meet demand, so ischemia appears with exertion and settles with rest.

· Vulnerable plaque features: a thin fibrous cap, a large lipid core, and many inflammatory cells.

· Macrophage metalloproteinases digest the collagen of the cap, and T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap is not repaired.

· Cap rupture exposes tissue factor and collagen, which trigger thrombosis, and a thrombus forms on the disrupted surface.

· This is why many myocardial infarctions arise from plaques that were not severely stenotic.

· Complications and mechanisms: stenosis from slow plaque growth limiting flow reserve; rupture or erosion with thrombosis from cap disruption exposing tissue factor and collagen; embolism when plaque debris or thrombus breaks off downstream; aneurysm because intimal plaque starves and weakens the underlying media; hemorrhage into plaque when its microvessels rupture and expand it abruptly.

Distinctions that matter

· Stenosis produces chronic demand ischemia; thrombosis on a disrupted plaque produces acute events.

· Plaque size predicts exertional symptoms, while plaque composition predicts sudden occlusion.

Common misconceptions

· Assuming the tightest stenosis is the most dangerous lesion. A modest plaque with a thin cap can occlude a vessel outright.

· Reading hemorrhage into a plaque as the same event as rupture of the cap. The first expands the plaque from within, the second exposes its contents to blood.

Where it is taught: Section C02, vulnerable plaque paragraph, complications table, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-RT01

Retrieval exercise: Rebuild the atherogenesis sequence

A competent response contains

· Step 1: endothelial dysfunction at branch points and curvatures, where flow is disturbed rather than laminar, which is why those sites and not straight segments.

· Step 2: LDL enters the intima and is oxidized.

· Step 3: monocytes adhere, migrate in, and become macrophages; the endothelium displays VCAM-1.

· Step 4: scavenger receptor uptake of oxidized LDL produces foam cells.

· Step 5: fatty streak, a flat intimal collection of foam cells; present in the aortas of virtually all children over about 10 and not obstructive.

· Step 6: PDGF and other growth factors recruit smooth muscle cells from the media.

· Step 7: a fibrous cap of collagen forms over a necrotic lipid core, and this mature plaque is the first lesion that meaningfully narrows the lumen.

· Step 8: rupture, erosion, hemorrhage, or calcification gives a complicated plaque.

Distinctions that matter

· The fatty streak is visible and early but does not obstruct; narrowing begins once smooth muscle and matrix build a raised plaque.

· VCAM-1 brings monocytes in; PDGF brings smooth muscle in. Different signals at different steps.

Common misconceptions

· Placing smooth muscle recruitment before foam cell formation. Foam cells and the fatty streak come first.

· Treating the fatty streak as a disease stage that must progress. It is near universal in children and only some streaks advance.

Where it is taught: Section C02, atherogenesis flow and the paragraph on fatty streaks of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-Q01

Board-style question: An infarct from a modest stenosis

Correct answer: C. Macrophage metalloproteinases degraded cap collagen while interferon gamma suppressed its replacement

Why it is right: The described plaque is vulnerable, and in such plaques macrophage metalloproteinases degrade the collagen of the cap while T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap tears, tissue factor and collagen are exposed, and the overlying thrombus occludes the artery.

A is wrong: Medial calcification, as in Monckeberg sclerosis, stiffens medium limb arteries without narrowing the lumen and plays no part in coronary plaque rupture.

B is wrong: Gradual plaque growth limiting flow reserve produces exertional ischemia once stenosis reaches about 70%, and at 40% it would not cause a resting occlusion or a fresh thrombus.

D is wrong: Plasma protein leakage with excess matrix is hyaline arteriolosclerosis of arterioles in hypertension and diabetes, not a mechanism of coronary cap rupture.

A competent response contains

· A thin fibrous cap, a large lipid core, and dense macrophage infiltration define a vulnerable plaque.

· Macrophage metalloproteinases digest cap collagen and T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap thins and is not repaired; rupture exposes tissue factor and collagen and a thrombus occludes the vessel.

Distinctions that matter

· Plaque stability rather than plaque size determines the risk of acute occlusion, which is why a 40% stenosis can cause a fatal infarct.

Common misconceptions

· Judging risk from the angiographic percentage alone.

Where it is taught: Section C02 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C03-BD1

Brain dump 1 · foundational retrieval: Draw LDL clearance and then break it

A competent response contains

· Hepatic LDL receptors bind apolipoprotein B-100 on LDL and remove it from plasma by clathrin-mediated endocytosis.

· Cholesterol released in the lysosome keeps SREBP-2 inactive in the endoplasmic reticulum, which lowers synthesis of HMG-CoA reductase and of new LDL receptors.

· LDLR: loss-of-function mutation in the receptor itself, the most common cause.

· APOB: loss-of-function mutation in the receptor binding site of the ligand.

· PCSK9: gain-of-function mutation in a protein that steers the receptor to lysosomal degradation, so fewer receptors return to the surface.

· Inheritance is autosomal dominant.

· Deposits are LDL-derived cholesterol taken up by macrophages: tendon xanthomas in the Achilles tendon and the extensor tendons of the hands; xanthelasma as yellow plaques on the eyelids; corneal arcus as a gray-white ring at the corneal margin.

· Tendon xanthomas are highly suggestive of the disease; xanthelasma and arcus are less specific because both occur in older adults with normal lipids, although arcus before about age 45 suggests it.

· Homozygotes may also have planar cutaneous xanthomas and lipid deposits in the aortic root and valve.

Distinctions that matter

· A loss-of-function mutation in LDLR or APOB and a gain-of-function mutation in PCSK9 all end in the same place: fewer working receptors clearing LDL.

· Tendon xanthoma is specific; eyelid and corneal deposits are common findings in older adults and carry less weight.

Common misconceptions

· Saying the liver overproduces cholesterol as the primary defect. The defect is clearance, and hepatic synthesis continues because the hepatocyte senses too little cholesterol inside itself.

· Treating PCSK9 as a receptor. It directs the receptor to lysosomal degradation.

Where it is taught: Section C03, first and third paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C03-BD2

Brain dump 2 · integration and application: Read the lipid panel and the tendons

A competent response contains

· The boy: LDL above 500 mg/dL with childhood xanthomas means little or no receptor activity, so he is homozygous, and coronary disease or infarction is expected in childhood or the teens.

· The mother: LDL above 190 mg/dL with a tendon xanthoma and early arcus means about half of normal receptor activity, so she is heterozygous, with coronary disease commonly in the fourth to fifth decade in men and somewhat later in women.

· Plasma LDL rises because clearance falls while the hepatocyte, sensing less intracellular cholesterol, keeps SREBP-2 active and continues synthesis; the lipid stays in plasma rather than entering the cell.

· Plasma LDL rises in proportion to the receptor defect, so gene dosage sets the level.

· Triglycerides are usually normal, which separates this disease from combined hyperlipidemias.

· Corneal arcus before about age 45 suggests the diagnosis, but in a 70-year-old it occurs with normal lipids and carries little weight.

· The same process makes all three lesions: LDL that is not cleared accumulates, macrophages take it up through scavenger receptors, and lipid-laden macrophages form the tendon nodule, the eyelid plaque, and the intimal plaque of premature atherosclerosis.

· Heterozygotes are about 1 in 250 people; homozygotes are rare.

Distinctions that matter

· Receptor activity, plasma LDL, and age at first coronary event all move together with gene dosage.

· Normal triglycerides with a very high LDL points at receptor-mediated clearance rather than at overproduction of triglyceride-rich particles.

Common misconceptions

· Expecting the liver to shut down cholesterol synthesis because plasma cholesterol is high. The hepatocyte responds to its own intracellular pool, not to plasma.

· Reading any corneal arcus as evidence of familial hypercholesterolemia regardless of age.

Where it is taught: Section C03, second and third paragraphs and the heterozygous against homozygous table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C04-BD1

Brain dump 1 · foundational retrieval: Build the aneurysm table from memory

A competent response contains

· An aneurysm is a localized permanent dilation of a vessel or the heart; a true aneurysm involves all three layers and a false aneurysm is a leak contained by adventitia or clot, often at an arterial puncture or graft anastomosis.

· An abdominal aorta of 3 cm or more is aneurysmal.

· Abdominal aortic: infrarenal aorta; atherosclerosis with inflammation and matrix metalloproteinases degrading the matrix; men over 60 who smoke, with hypertension and family history as added risks; rupture, with risk rising steeply with diameter.

· Thoracic aortic: ascending aorta and arch; cystic medial degeneration from hypertension or connective tissue disorders such as Marfan syndrome, and tertiary syphilis; aortic root dilation with regurgitation, compression of mediastinal structures, and dissection.

· Berry or saccular: circle of Willis, mostly anterior circulation branch points; thin or absent media at branch points, with ADPKD, Ehlers-Danlos syndrome, hypertension, and smoking; subarachnoid hemorrhage.

· Mycotic: any artery, often aorta or cerebral branches; septic emboli from infective endocarditis or bacteremic seeding of plaque, for example Salmonella; rapid growth and rupture.

· Wall stress is about pressure times radius divided by wall thickness, the law of Laplace.

· Once a segment dilates, the larger radius raises stress, the stretched wall thins which raises it further, and higher stress degrades the remaining matrix, so dilation tends to continue.

· The infrarenal media has few vasa vasorum, so it depends on diffusion from the lumen, and a thick plaque impairs that diffusion and starves the media.

Distinctions that matter

· A false aneurysm is a contained leak rather than a dilated wall, so it has no intact media of its own.

· An aneurysm always means the media has failed; what differs between types is why it failed.

Common misconceptions

· Expecting abdominal aneurysms above the renal arteries. The usual site is infrarenal.

· Treating a berry aneurysm as atherosclerotic. It arises from a congenitally thin or absent media at a branch point.

Where it is taught: Section C04, first paragraph, aneurysm types table, and the worked example on wall stress of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C04-BD2

Brain dump 2 · integration and application: Work the numbers on a growing aorta

A competent response contains

· Radius goes from 1.0 cm to 3.0 cm, so at the same pressure and wall thickness, stress is about three times greater.

· As the wall stretches it thins, and thickness is in the denominator, so stress rises further than the radius change alone predicts.

· Higher stress degrades the remaining matrix, allowing more dilation, which is why expansion accelerates with diameter and rupture risk climbs steeply above about 5.5 cm; risk is very low below 4 cm.

· Hypertension raises the pressure term, so it adds to every step of the feedback.

· Rupture presents with abdominal or back pain, hypotension, and a pulsatile abdominal mass.

· The young tall patient has cystic medial degeneration, which is loss of elastic fibers and smooth muscle with pools of ground substance in the media, as in Marfan syndrome.

· Dilation of the root pulls the valve commissures apart so the cusps no longer meet, giving aortic regurgitation and a diastolic murmur.

· Tertiary syphilis causes obliterative endarteritis of the vasa vasorum of the ascending aorta, so the media becomes ischemic; the intima has a wrinkled tree-bark appearance and the root dilates with regurgitation.

Distinctions that matter

· Marfan syndrome weakens the media directly through defective matrix, while syphilis weakens it by cutting off its blood supply, and both end in root dilation.

· Thoracic and abdominal aneurysms have different usual mechanisms, so the site points to the process.

Common misconceptions

· Calculating stress from diameter instead of radius, which gives the right ratio by luck but the wrong quantity.

· Assuming the aortic valve itself is diseased when the root dilates. The cusps are normal and simply fail to meet.

Where it is taught: Section C04, worked example on wall stress and the paragraphs on abdominal and thoracic aneurysms of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild a dissection and its consequences

A competent response contains

· In aortic dissection blood enters the media through an intimal tear and splits the wall along its length, creating a false lumen.

· The underlying problem is a weakened media. Hypertension is present in most patients and causes medial degeneration while raising the shear and wall stress that drive the split.

· Younger patients usually have a connective tissue or structural defect: cystic medial degeneration in Marfan syndrome with defective fibrillin-1 and excess TGF-beta signaling, Ehlers-Danlos syndrome, bicuspid aortic valve with its aortopathy, or Turner syndrome, often with bicuspid valve and coarctation.

· Acute triggers are abrupt pressure surges, as with cocaine or the hemodynamic load of pregnancy.

· Sequence: weakened media plus high wall stress; intimal tear, usually in the ascending aorta; blood splits the media into a false lumen; proximal spread to the aortic root with loss of valve support and regurgitation; rupture into the pericardium with hemopericardium and tamponade; branch compression causing coronary, carotid, subclavian, renal, or mesenteric ischemia; external rupture with hemothorax or exsanguination.

· Outward rupture, into the pericardium, pleural cavity, or peritoneum, is the usual cause of death.

· Stanford A involves the ascending aorta, with or without the descending aorta. Stanford B does not involve the ascending aorta and usually begins beyond the left subclavian artery.

· Heavy atherosclerosis or medial scarring tends to stop a dissection from spreading, so dissection is not a typical complication of atherosclerosis.

Distinctions that matter

· A dissection splits the media while an aneurysm dilates it; both start from a weakened media and can occur in the same aorta.

· The ascending aorta decides the Stanford type because its involvement threatens the valve, the pericardium, and the coronary arteries.

Common misconceptions

· Listing dissection as a complication of atherosclerosis. Plaque and scarring tend to arrest a dissection.

· Describing the false lumen as a rupture through the whole wall. The blood stays within the media until it ruptures outward.

Where it is taught: Section C05, first paragraph, the dissection flow, and the Stanford table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-BD2

Brain dump 2 · integration and application: Explain each sign in a dissecting aorta

A competent response contains

· Tearing pain moving to between the scapulae reflects the dissection advancing distally along the media, with pain tracking the splitting wall.

· Unequal arm pressures arise when the false lumen narrows a subclavian artery, so the pulse is weak and the pressure is lower on that side.

· A new diastolic murmur is aortic regurgitation from proximal spread into the root with loss of valve support.

· A widened mediastinum on the chest radiograph reflects the dilated, blood-filled aorta and surrounding hematoma.

· Hypotension with distended neck veins and muffled heart sounds means rupture into the pericardium; blood in the pericardial sac raises intrapericardial pressure above filling pressures, so ventricular filling falls and cardiac output falls, which is tamponade.

· Tamponade and aortic regurgitation both require the ascending aorta, so this is Stanford A, the type with the highest risk along with coronary occlusion.

· A tear beyond the left subclavian artery is Stanford B, typical of older patients with hypertension, with lower risk and complications mainly branch ischemia and rupture.

· Branch occlusion shows as neurologic deficits or stroke for the carotid, paraplegia for the spinal supply, oliguria for the renal arteries, and abdominal pain for the mesenteric arteries.

Distinctions that matter

· Unequal arm pressures appear in both dissection and Takayasu arteritis; sudden tearing pain is what points to dissection.

· Hypotension in dissection can come from tamponade or from external rupture and blood loss, and the neck veins separate them.

Common misconceptions

· Reading hypotension with distended neck veins as heart failure. In this setting it is impaired filling from pericardial blood.

· Expecting the pain to stay in one place. It moves as the dissection advances.

Where it is taught: Section C05, clinical paragraph, the dissection flow, Stanford table, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-Q02

Board-style question: Tearing chest pain in a tall young man

Correct answer: A. Blood has ruptured from the aorta into the pericardial sac and is limiting ventricular filling

Why it is right: A Stanford A dissection in a patient with a connective tissue disorder can rupture back into the pericardial sac, and blood there raises intrapericardial pressure above filling pressures so ventricular filling and cardiac output fall, which produces hypotension with distended neck veins and muffled heart sounds.

B is wrong: Renal artery occlusion by the false lumen causes oliguria and renin-mediated hypertension rather than acute hypotension, and it does not distend the neck veins or muffle the heart sounds.

C is wrong: Fibrinoid necrosis of arterioles belongs to hyperplastic arteriolosclerosis in severe hypertension and causes renal ischemia, not acute tamponade.

D is wrong: An intimal plaque starving the media produces an aneurysm over years, and heavy atherosclerosis actually tends to stop a dissection from spreading.

A competent response contains

· Tall stature with long fingers and a high arched palate suggests Marfan syndrome with cystic medial degeneration, and sudden tearing chest pain radiating to the back is aortic dissection.

· Hypotension with distended jugular veins and muffled heart sounds means blood has ruptured into the pericardium, so intrapericardial pressure limits ventricular filling.

Distinctions that matter

· Hypotension in dissection can come from tamponade or from external rupture with blood loss, and distended neck veins point to tamponade.

Common misconceptions

· Reading raised jugular pressure in this setting as volume overload.

Where it is taught: Section C05 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-BD1

Brain dump 1 · foundational retrieval: Claudication, the ankle-brachial index, and the cholesterol shower

A competent response contains

· Peripheral arterial disease is atherosclerotic narrowing of the arteries to the legs, most strongly linked to smoking and diabetes.

· At rest a stenosed artery still delivers enough blood; during walking the calf arterioles dilate and demand rises, but the fixed stenosis prevents flow from increasing, so the muscle becomes ischemic.

· Intermittent claudication is cramping pain in the muscle group below the stenosis, reproduced by a set walking distance and relieved within minutes by rest.

· Superficial femoral disease causes calf pain; aortoiliac disease causes buttock and thigh pain.

· Index bands: above 1.40 means noncompressible calcified arteries and an unreliable result; 1.00 to 1.40 normal; 0.91 to 0.99 borderline; 0.90 or lower confirms disease because pressure is lost across stenoses; below about 0.40 severe disease, often with rest pain or tissue loss.

· Normally ankle pressure equals or slightly exceeds arm pressure because of pulse wave amplification, so a lower ankle pressure means flow has crossed a resistance.

· Chronic ischemia gives weak or absent pulses, cool skin, hair loss, thin shiny skin, thickened nails, and dependent rubor with pallor on elevation.

· Rest pain in the forefoot, typically at night, signals critical ischemia; hanging the foot down adds gravity to perfusion pressure, which raises flow and eases the pain. Nonhealing ulcers on the toes and pressure points and gangrene follow.

· Leriche syndrome is occlusion at the aortic bifurcation with buttock and thigh claudication, absent or diminished femoral pulses, and erectile dysfunction from reduced internal iliac flow.

· Atheroembolic disease: an ulcerated plaque in the aorta or a large artery showers cholesterol crystals downstream, classically after arterial catheterization, angiography, or vascular surgery, and occasionally with anticoagulation or thrombolysis.

· Crystals lodge in arterioles of about 100 to 200 micrometers and incite inflammation, giving blue or purple toes and livedo reticularis with intact pedal pulses because the large arteries remain open; the kidney shows progressive renal failure, and blood tests may show eosinophilia and low complement.

· Biopsy shows biconvex, needle-shaped clefts in arterioles where cholesterol crystals dissolved during tissue processing.

Distinctions that matter

· Claudication is a flow reserve problem and rest pain is a resting perfusion problem, which is why one comes with walking and the other at night.

· An index above 1.40 and an index of 0.90 or lower both signal arterial disease, but only the low value measures lost pressure.

Common misconceptions

· Expecting absent pulses in cholesterol embolization. The large arteries stay open, so the pulses are intact.

· Treating elevation as relief for ischemic rest pain. Elevation lowers perfusion pressure and makes arterial pain worse.

Where it is taught: Section C06, all paragraphs and the ankle-brachial index table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-BD2

Brain dump 2 · integration and application: Three painful feet, three different vessels

A competent response contains

· Patient 1: atherosclerotic peripheral arterial disease of the superficial femoral artery, a large to medium artery obstructed by intimal plaque, so pressure is lost across the stenosis and the pedal pulses are absent, with an index of 0.58 confirming disease.

· Patient 2: atheroembolic or cholesterol embolization disease after angiography, with cholesterol crystals in arterioles of about 100 to 200 micrometers, so the small vessels are blocked while the large arteries remain open and the pedal pulses stay normal.

· Patient 2's blue toes, livedo reticularis, rising creatinine, eosinophilia, and low complement all fit crystal-induced inflammation of small vessels.

· Patient 3: Monckeberg medial calcific sclerosis makes the arteries noncompressible, so the index of 1.46 is uninterpretable and does not exclude coexisting atherosclerotic disease, which the nonhealing toe ulcer suggests is present.

· In patient 3 use toe pressures and the toe-brachial index, pulse volume recordings or Doppler waveforms, or imaging, because the digital arteries are less often calcified.

· The level of obstruction explains the pulses: proximal large artery disease abolishes the pulse, distal arteriolar embolism does not.

· Patient 1 and patient 3 have arterial ulcers, which are punched out, dry, and pale and sit at the toes, heel, and pressure points, because perfusion pressure is too low to supply the tissue, unlike the shallow moist ulcers of venous disease above the medial malleolus.

Distinctions that matter

· Blue toes with normal pulses point to small vessel embolism; blue or ulcerated toes with absent pulses point to large artery stenosis.

· A normal or high index never rules out disease in a calcified limb, whereas a low index is reliable evidence of it.

Common misconceptions

· Reading eosinophilia in patient 2 as an allergy to contrast. It is part of the inflammatory response to cholesterol crystals.

· Trusting an index above 1.40 in a patient with diabetes or chronic kidney disease.

Where it is taught: Section C06, paragraphs on peripheral arterial disease and atheroembolic disease and the index table, with Monckeberg sclerosis in C01 and ulcer features in C09 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-PR01

Prediction exercise: Predict what happens as a femoral stenosis tightens

A competent response contains

· Resting calf flow stays adequate while the stenosis is moderate, because resting demand is low and a stenosed artery can still meet it.

· During walking the arterioles dilate and demand rises, but the fixed stenosis prevents flow from increasing, so the muscle becomes ischemic and he feels cramping calf pain that eases within minutes of rest.

· Ankle systolic pressure falls because pressure is lost across the stenosis, so the index falls below 0.90 and keeps falling as the narrowing tightens.

· Claudication distance shortens, because less of the flow reserve remains, so ischemia is reached sooner.

· Once resting flow is inadequate, forefoot pain appears at rest and typically at night; hanging the foot down adds a gravitational column to perfusion pressure, which raises flow and eases the pain.

· The leg loses hair, the skin becomes thin, shiny, and cool, the nails thicken, dependent rubor with pallor on elevation appears, and nonhealing ulcers form at the toes and pressure points, with gangrene if ischemia continues.

Distinctions that matter

· Flow reserve, not resting flow, is what a moderate stenosis consumes, which is why symptoms begin on exertion.

· Rest pain and claudication differ in whether resting perfusion is still sufficient, not in the kind of pain.

Common misconceptions

· Expecting resting flow to fall as soon as a stenosis appears. Resting flow is preserved until the narrowing is severe.

· Explaining relief on dangling the foot by reduced venous pressure. The mechanism is added gravitational perfusion pressure.

Where it is taught: Section C06, first and second paragraphs and the index table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-BD1

Brain dump 1 · foundational retrieval: Rebuild pulmonary embolism from clot to blood gas

A competent response contains

· More than 95% of pulmonary emboli arise from thrombi in the deep veins of the leg, especially the proximal popliteal, femoral, and iliac veins, because those veins are large enough to release sizable clots.

· Sequence: embolus obstructs pulmonary arteries; pulmonary vascular resistance rises abruptly; the right ventricle dilates and fails, which is acute cor pulmonale; blood is diverted to open lung units creating low ventilation to perfusion areas; hypoxemia with a widened alveolar-arterial gradient; hypoxemia and receptor stimulation drive hyperventilation; low PaCO2 gives respiratory alkalosis.

· In the obstructed region ventilation continues without perfusion, which is dead space; in the open region perfusion rises relative to ventilation, so blood leaves it less fully oxygenated; local loss of surfactant and atelectasis add shunt.

· Obstruction of roughly 60% or more of the pulmonary circulation can cause shock or sudden death, and a large saddle embolus at the bifurcation of the main pulmonary artery can cause sudden death.

· Large embolus: sudden dyspnea with tachypnea and tachycardia, hypotension and shock in massive cases.

· Small peripheral embolus: pleuritic chest pain and hemoptysis, from pleural irritation or infarction.

· Right ventricular strain: dilated right ventricle, raised jugular venous pressure, sinus tachycardia as the most common electrocardiographic finding, sometimes a new right bundle branch block or an S1Q3T3 pattern, and elevated troponin or BNP.

· D-dimer is a fibrin degradation product that is sensitive but not specific, so a raised value does not establish the diagnosis.

Distinctions that matter

· Calf vein thrombi are common but proximal thrombi supply the clinically important emboli.

· Dead space and the overperfused open units are two halves of the same mismatch, and both are needed to explain the hypoxemia.

Common misconceptions

· Expecting PaCO2 to rise because dead space has increased. The patient hyperventilates, so PaCO2 falls.

· Treating a raised D-dimer as diagnostic. It is sensitive and not specific.

Where it is taught: Section C07, first and third paragraphs and the pulmonary embolism flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-BD2

Brain dump 2 · integration and application: Calculate the gradient and explain the lung that lives

A competent response contains

· Alveolar PO2 equals 0.21 times (760 minus 47) minus PaCO2 divided by 0.8, which is 0.21 times 713, about 150, minus 28 divided by 0.8, which is 35, giving about 115 mm Hg.

· The gradient is 115 minus 58, about 57 mm Hg, far above the normal of roughly 5 to 15 mm Hg in a young adult.

· A PaCO2 of 28 mm Hg shows she is hyperventilating, and hypoventilation requires a raised PaCO2, so hypoventilation cannot explain the hypoxemia.

· The mechanism is ventilation-perfusion mismatch, with shunt from surfactant loss and atelectasis adding to it.

· Only about 10% of emboli cause infarction because the lung has a dual supply: bronchial arteries from the systemic circulation continue to perfuse tissue distal to the block, and alveolar oxygen diffuses into the tissue directly.

· Infarction occurs mainly when bronchial flow is already compromised, as in left heart failure, where pulmonary venous pressure is also high.

· The infarct is hemorrhagic, wedge-shaped, and based on the pleura.

· Repeated small emboli over time produce chronic thromboembolic pulmonary hypertension.

Distinctions that matter

· A widened gradient means a gas exchange defect, while a normal gradient with hypoxemia and a high PaCO2 means hypoventilation.

· Occlusion of a pulmonary artery is not the same as loss of blood supply to lung tissue, because of the bronchial circulation.

Common misconceptions

· Forgetting to subtract water vapour pressure of 47 mm Hg before applying the fraction of inspired oxygen.

· Assuming every pulmonary artery occlusion infarcts the lung. Most do not.

Where it is taught: Section C07, third and fourth paragraphs, the worked example on the gradient, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-RT02

Retrieval exercise: Rebuild the pulmonary embolism sequence

A competent response contains

· Step 1: embolus obstructs pulmonary arteries.

· Step 2: pulmonary vascular resistance rises abruptly; the right ventricle fails because it is thin walled and built for a low-pressure circuit, and acute failure from a sudden pressure load is acute cor pulmonale.

· Step 3: the right ventricle dilates and fails.

· Step 4: blood is diverted to open lung units, creating low ventilation to perfusion areas; the obstructed unit is ventilated and not perfused, which is dead space, and the open unit is perfused in excess of its ventilation.

· Step 5: hypoxemia with a widened alveolar-arterial gradient, which separates a gas exchange defect from hypoventilation.

· Step 6: hypoxemia and receptor stimulation drive hyperventilation, so respiratory rate and minute ventilation rise.

· Step 7: low PaCO2 produces respiratory alkalosis.

· Obstruction of roughly 60% or more of the pulmonary circulation can cause shock or sudden death.

Distinctions that matter

· Dead space raises the work of breathing but the arterial hypoxemia comes from the overperfused open units.

· The left ventricle is not the chamber at risk here, because the pressure load lies upstream of it.

Common misconceptions

· Drawing the obstructed unit as unventilated. It is ventilated and not perfused.

· Expecting the arterial blood gas to show carbon dioxide retention.

Where it is taught: Section C07, pulmonary embolism flow and third paragraph of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-CA01

Clinical application: Sudden breathlessness after a long flight

A competent response contains

· Recognize: pulmonary embolism from a left proximal leg deep vein thrombosis, with stasis from a 14-hour flight, hypercoagulability from estrogen, and endothelial or venous wall involvement in the already thrombosed calf and thigh veins, so all three elements of Virchow triad are present.

· Explain: the embolus creates regions that are ventilated but not perfused while blood is diverted to open units that become overperfused relative to their ventilation, so blood leaving them is not fully oxygenated and arterial PO2 falls; surfactant loss and atelectasis add shunt; hypoxemia and receptor stimulation drive hyperventilation, which lowers PaCO2.

· Differentiate: pure hypoventilation raises PaCO2 and leaves the alveolar-arterial gradient normal, whereas here PaCO2 is low and the gradient is wide; a right-to-left shunt also widens the gradient but the hypoxemia corrects poorly with supplemental oxygen, while ventilation-perfusion mismatch corrects readily.

· Investigate: alveolar PO2 is 0.21 times (760 minus 47) minus 29 divided by 0.8, which is about 150 minus 36, or about 114 mm Hg; the gradient is 114 minus 56, about 58 mm Hg, far above the normal 5 to 15 mm Hg, confirming a gas exchange defect; compression ultrasound of the left leg would show a vein that cannot be compressed because clot fills its lumen.

· Decide: the right ventricle is under acute strain, shown by the raised jugular venous pressure, the loud pulmonary component of the second heart sound, and the sinus tachycardia, which is the most common electrocardiographic finding.

· Justify: bronchial arteries from the systemic circulation continue to perfuse tissue distal to the obstruction and alveolar oxygen diffuses into the tissue directly, so only about 10% of emboli infarct the lung; left heart failure, which reduces bronchial flow and raises pulmonary venous pressure, would make a hemorrhagic wedge-shaped pleura-based infarct likely instead.

Distinctions that matter

· Pleuritic pain and a clear chest radiograph fit a peripheral embolus without infarction, so pain does not prove tissue death.

· A normal chest radiograph with marked hypoxemia is itself informative, because parenchymal causes would usually show something.

Common misconceptions

· Reading the low PaCO2 as evidence against a serious problem. It is the expected consequence of hyperventilation.

· Assuming a swollen calf means the embolus came from the calf veins. The clinically important emboli come from the proximal veins.

Where it is taught: Section C07, all paragraphs, the flow, and the worked example, with Virchow triad in C08 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-BD1

Brain dump 1 · foundational retrieval: Build Virchow triad and the venous thrombus

A competent response contains

· Venous thrombi usually begin in the valve pockets of the deep calf veins, where flow is slowest, and can propagate proximally.

· They form in low-flow, low-pressure vessels and are rich in red cells and fibrin, which makes them red thrombi, unlike platelet-rich arterial thrombi.

· Stasis: slow flow lets activated factors accumulate and keeps platelets near the endothelium. Examples include immobility, long travel, paralysis, heart failure, and compression by the pregnant uterus.

· Endothelial injury: loss of the anticoagulant surface and exposure of tissue factor. Examples include surgery and trauma, especially hip and knee, central venous catheters, and prior deep vein thrombosis.

· Hypercoagulability: excess procoagulant or loss of natural anticoagulant. Examples include factor V Leiden, prothrombin G20210A, antiphospholipid antibodies, malignancy, estrogen and pregnancy, and nephrotic syndrome through urinary antithrombin loss.

· Major surgery and cancer are high risk because they combine all three elements.

· Nephrotic syndrome works by losing antithrombin in the urine.

· A thrombosis obstructs venous outflow, raising capillary hydrostatic pressure, while inflammation of the vein wall adds pain and warmth, giving one swollen, warm, tender leg, sometimes with dilated superficial veins and a larger calf circumference than the other side.

· Many are silent because collateral veins drain the limb, which is why pulmonary embolism may be the first sign.

· Calf pain on dorsiflexion, the Homans sign, is unreliable.

· On ultrasound the thrombosed vein cannot be compressed because clot fills its lumen.

Distinctions that matter

· Red venous thrombi and white arterial thrombi differ because flow conditions decide whether red cells and fibrin or platelets dominate.

· Swelling comes from obstructed outflow and raised capillary pressure, while pain and warmth come from inflammation of the vein wall.

Common misconceptions

· Expecting every deep vein thrombosis to produce a swollen leg. Collaterals keep many of them silent.

· Relying on the Homans sign, which the notes state is unreliable.

Where it is taught: Section C08, first and third paragraphs and the Virchow triad table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-BD2

Brain dump 2 · integration and application: A red cord, a swollen leg, and a wandering vein

A competent response contains

· Patient 1: superficial thrombophlebitis of a varicose vein, in the superficial venous system, with low embolic risk unless the clot extends into the deep system.

· Patient 2: deep vein thrombosis of the left leg, in the deep system of calf, popliteal, femoral, or iliac veins, with significant embolic risk that is highest with proximal clot.

· Patient 3: migratory thrombophlebitis, the Trousseau sign of malignancy, which is recurrent superficial and deep venous thrombosis appearing and resolving at different sites.

· Patient 2 has diffuse swelling because the deep veins drain most of the leg, so their occlusion raises capillary hydrostatic pressure throughout the limb; patient 1 has little swelling because deep drainage is intact.

· The responsible tumours are adenocarcinomas, classically pancreatic and also lung, gastric, and colon; painless jaundice with weight loss points first to the pancreas.

· The tumour releases tissue factor and mucin, which activate coagulation.

· The Trousseau sign of hypocalcemia is a different finding: carpal spasm under a blood pressure cuff.

· All three patients would have a normal platelet count, because these are thrombotic rather than thrombocytopenic conditions, so the count does not separate them.

Distinctions that matter

· A palpable cord with no swelling means a superficial vein; diffuse swelling of the whole limb means the deep system.

· Two different signs carry the Trousseau name, and only one concerns veins and cancer.

Common misconceptions

· Treating superficial thrombophlebitis as a source of pulmonary embolism. It rarely embolizes unless it extends into the deep veins.

· Reading recurrent thrombophlebitis at varied sites as bad luck rather than as a sign that prompts a search for a visceral adenocarcinoma.

Where it is taught: Section C08, third and fourth paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-Q03

Board-style question: Thrombosis at shifting sites

Correct answer: D. Tissue factor and mucin released by an adenocarcinoma activating coagulation

Why it is right: Visceral adenocarcinomas, classically pancreatic, release tissue factor and mucin that activate coagulation, producing recurrent superficial and deep venous thrombosis that appears and resolves at different sites, and painless jaundice with marked weight loss fits a pancreatic primary.

A is wrong: Antiphospholipid antibodies cause thrombosis but not this migratory pattern with painless jaundice and weight loss, and they typically prolong the activated partial thromboplastin time.

B is wrong: Urinary antithrombin loss in nephrotic syndrome causes thrombosis in the setting of heavy proteinuria and edema, which is not the picture here.

C is wrong: Compression of the deep veins by a pelvic mass produces persistent obstruction of the affected limb rather than cords that resolve in one site and appear in another, including the forearm.

A competent response contains

· Recurrent superficial and deep thrombosis appearing and resolving at different sites is migratory thrombophlebitis, the Trousseau sign of malignancy.

· It occurs with adenocarcinomas, classically pancreatic and also lung, gastric, and colon, which release tissue factor and mucin that activate coagulation; weight loss with painless jaundice points to the pancreas.

Distinctions that matter

· The Trousseau sign of malignancy is migratory venous thrombosis, while the Trousseau sign of hypocalcemia is carpal spasm under a blood pressure cuff.

Common misconceptions

· Expecting an abnormal platelet count or coagulation screen to mark a hypercoagulable state.

Where it is taught: Section C08 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-BD1

Brain dump 1 · foundational retrieval: The calf pump, the valves, and the skin that follows

A competent response contains

· Leg veins return blood against gravity using the calf muscle pump and one-way valves.

· During walking, calf contraction squeezes the deep veins, and competent valves let blood move only upward and from superficial to deep through the perforating veins, which keeps ankle venous pressure low during movement.

· When valves fail, the column of blood is no longer interrupted and pressure in the superficial veins stays high even with walking, which is ambulatory venous hypertension.

· Varicose veins are dilated, tortuous superficial veins, most often in the great saphenous system, produced by raised intraluminal pressure and weakness of the vein wall that make the valves incompetent.

· Risk rises with age, female sex, pregnancy, obesity, prolonged standing, and family history; pregnancy adds compression of the inferior vena cava by the uterus plus hormonal relaxation of the vein wall, and prolonged standing keeps the hydrostatic column uninterrupted.

· Prior deep vein thrombosis that damages deep valves causes secondary venous insufficiency, and superficial varicose veins rarely cause embolism.

· Portal hypertension raises pressure elsewhere and produces esophageal varices and hemorrhoids.

· Sequence: valve incompetence in superficial, perforating, or deep veins; sustained venous hypertension in the lower leg; raised capillary hydrostatic pressure giving edema; red cells leak and hemosiderin stains the skin brown; leukocyte trapping and pericapillary fibrin impair tissue oxygenation; stasis dermatitis and fibrosis, which is lipodermatosclerosis; minor trauma fails to heal, giving a venous ulcer.

· The brown colour is hemosiderin, iron released from leaked red cells.

Distinctions that matter

· Valves and the calf pump work together: the pump moves blood and the valves stop it coming back, so failure of either raises ankle pressure.

· Primary varicose disease starts in the vein wall and its valves, while secondary insufficiency follows damage to deep valves by thrombosis.

Common misconceptions

· Expecting standing venous pressure to be the problem. The abnormality is that pressure stays high during walking, when it should fall.

· Treating varicose veins as a source of pulmonary embolism.

Where it is taught: Section C09, first and second paragraphs and the venous ulcer flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-BD2

Brain dump 2 · integration and application: Two ulcers above two different problems

A competent response contains

· The woman has chronic venous insufficiency with a venous ulcer: location above the medial malleolus in the gaiter area, where pressure from incompetent perforators and the great saphenous system is highest; shallow, irregular, moist with a granulating base; mild to moderate pain eased by elevation; surrounding edema, brown pigment, and stasis dermatitis.

· The man has an arterial ulcer: at the toes, heel, pressure points, and distal foot; punched out, dry, with a pale base; severe pain that is worse with elevation; surrounding skin cool, shiny, and hairless with weak pulses.

· Elevation lowers venous pressure, which reduces capillary hydrostatic pressure and edema and helps the venous ulcer; elevation also lowers arterial perfusion pressure, which worsens ischemia and arterial pain.

· Evening edema: raised capillary hydrostatic pressure from sustained venous hypertension drives fluid into the tissue through the day, and overnight recumbency removes the hydrostatic column so the fluid is reabsorbed.

· Brown discoloration: red cells leak from congested capillaries and hemosiderin from their iron stains the skin.

· Itchy scaling skin: stasis dermatitis from chronic edema and inflammation.

· Indurated tissue: lipodermatosclerosis, fibrosis of dermis and subcutaneous fat after prolonged inflammation.

· The ulcer itself: leukocyte trapping and pericapillary fibrin impair oxygen delivery, so minor trauma fails to heal.

Distinctions that matter

· Both ulcers sit on the lower leg, and location, moisture, and the response to elevation separate them.

· Pain that improves with elevation points to venous pressure, and pain that worsens with elevation points to arterial inflow.

Common misconceptions

· Elevating the leg of a patient with an ischemic foot. It lowers perfusion pressure and increases pain.

· Reading brown pigmentation as bruising or hyperpigmentation of aging rather than as hemosiderin from chronic venous hypertension.

Where it is taught: Section C09, third paragraph, the venous ulcer flow, and the venous against arterial ulcer table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-RT03

Retrieval exercise: Rebuild the path from a failed valve to a broken skin surface

A competent response contains

· Step 1: valve incompetence in superficial, perforating, or deep veins; a previous deep vein thrombosis damages deep valves and causes secondary insufficiency.

· Step 2: sustained venous hypertension in the lower leg.

· Step 3: raised capillary hydrostatic pressure gives edema; the pressure that has risen is hydrostatic pressure in the capillary.

· Step 4: red cells leak and hemosiderin stains the skin brown; the iron from leaked red cells is the pigment.

· Step 5: leukocyte trapping and pericapillary fibrin impair tissue oxygenation, which are the two changes that reduce oxygen delivery.

· Step 6: stasis dermatitis, then fibrosis, which is lipodermatosclerosis; this is the step at which the changes become a visible skin disease.

· Step 7: minor trauma fails to heal, giving a venous ulcer, most often just above the medial malleolus.

Distinctions that matter

· Edema comes from pressure, while the pigment comes from red cells escaping, so the two findings reflect different consequences of the same hypertension.

· Stasis dermatitis is inflammatory and reversible in appearance, while lipodermatosclerosis is fibrotic and persistent.

Common misconceptions

· Placing the ulcer directly after edema and skipping the inflammatory and fibrotic steps that make the skin fragile.

· Attributing poor healing to edema alone. Leukocyte trapping and pericapillary fibrin impair oxygen delivery.

Where it is taught: Section C09, venous ulcer flow and the surrounding paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C10-BD1

Brain dump 1 · foundational retrieval: The three colours of a vasospastic attack

A competent response contains

· Raynaud phenomenon is episodic, reversible vasospasm of the small arteries and arterioles of the fingers and toes, and sometimes the nose and ears, triggered by cold or emotional stress.

· Digital vessels normally constrict to cold through sympathetic alpha-adrenergic signaling, mainly alpha-2 receptors, in order to conserve core heat.

· In Raynaud phenomenon that response is exaggerated and the lumen closes completely.

· Sequence: cold or stress causes intense digital artery spasm; white pallor from absent arterial inflow; blue cyanosis as stagnant blood is deoxygenated; spasm relaxes on rewarming; red reactive hyperemia with throbbing and tingling.

· White means no inflow, blue means oxygen has been extracted from blood trapped in the digit, and red means inflow has returned above baseline.

· The red phase is reactive hyperemia: during ischemia local metabolites including adenosine, carbon dioxide, hydrogen ions, and potassium accumulate and dilate the arterioles, so flow rebounds well above normal when the spasm lifts.

· Pallor is the most specific colour, and not every attack shows all three.

Distinctions that matter

· The blue phase is not a new episode of obstruction; it is the same trapped blood losing its oxygen.

· Normal cold-induced vasoconstriction and Raynaud phenomenon differ in degree, with complete closure of the lumen in the latter.

Common misconceptions

· Expecting all three colours in every attack, which leads to dismissing a real case.

· Explaining the red phase as inflammation. It is metabolite-driven reactive hyperemia.

Where it is taught: Section C10, first paragraph, the colour sequence flow, and the reactive hyperemia paragraph of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C10-BD2

Brain dump 2 · integration and application: Which Raynaud is in front of you

A competent response contains

· Patient 1: primary Raynaud phenomenon, a functional disorder of vascular reactivity with no underlying disease, with a structurally normal vessel between attacks and no tissue damage expected.

· Patient 2: secondary Raynaud phenomenon, with structural or inflammatory narrowing of the vessel, so spasm on an already narrow lumen can cause ischemic injury with digital pits, ulcers, and possible gangrene.

· Between attacks the primary patient's lumen is normal, so even complete spasm is brief and fully reversible; the secondary patient starts from a narrowed lumen, so the same spasm produces critical ischemia.

· Patient 2's tight shiny skin, asymmetric attacks, digital pits, and abnormal nailfold capillaries point to systemic sclerosis, in which intimal proliferation narrows the digital arteries.

· Other causes of secondary disease: lupus, thromboangiitis obliterans, atherosclerosis, cryoglobulinemia, and repetitive use of vibrating tools.

· Features that suggest an underlying disease: onset after about 30 to 40, asymmetric attacks, digital pits or ulcers, abnormal dilated or dropped-out nailfold capillaries, and positive autoantibodies.

· Primary disease is most common in young women, is usually symmetric, and has normal nailfold capillaries with a negative antinuclear antibody.

Distinctions that matter

· Tissue damage implies a structurally narrowed vessel, which points to secondary disease.

· Symmetry and age at onset are useful because a functional disorder affects both hands alike from a young age.

Common misconceptions

· Treating a positive antinuclear antibody as necessary for secondary disease. Nailfold changes and tissue loss count on their own.

· Assuming any patient with colour change has a connective tissue disease. Primary disease is common and benign.

Where it is taught: Section C10, fourth paragraph and the primary against secondary table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C11-BD1

Brain dump 1 · foundational retrieval: Thromboangiitis obliterans from histology to toes

A competent response contains

· Thromboangiitis obliterans is a segmental, thrombosing, acute and chronic inflammation of medium and small arteries, mainly the tibial and radial arteries and their digital branches, that spreads secondarily to adjacent veins and nerves.

· It occurs almost exclusively in heavy smokers, classically begins before about age 35 to 40, and is more often in men.

· The mechanism is thought to involve hypersensitivity or direct endothelial toxicity from tobacco products, and the disease rarely progresses once all tobacco use stops.

· Microscopically the involved segment shows a luminal thrombus containing microabscesses, which are small collections of neutrophils surrounded by granulomatous inflammation, with the internal elastic lamina relatively preserved.

· Over time the thrombus organizes and recanalizes, and the artery, vein, and nerve become bound together in fibrous tissue.

· Segmental means the lesions are sharply demarcated with normal vessel between them, which separates the disease from the diffuse plaques of atherosclerosis.

· Clinical order: cold sensitivity and Raynaud phenomenon, and nodular superficial phlebitis migrating along veins; then claudication of the instep of the foot or the forearm and hand, because the disease is distal; then intense rest pain from inflamed nerves, out of proportion to the visible lesion; then painful digital ulcers and gangrene, often needing amputation if smoking continues.

· Peripheral pulses at the wrist and ankle may be absent while proximal pulses are normal.

Distinctions that matter

· Preserved internal elastic lamina and neutrophil microabscesses within the thrombus separate this lesion from both atherosclerosis and most vasculitides.

· Pain here has two sources, ischemia and inflamed nerves, which is why it exceeds what the lesion looks like.

Common misconceptions

· Expecting proximal pulses to be lost. The disease is distal, so brachial and femoral pulses are normal.

· Treating the vein and nerve involvement as a separate problem. The inflammation spreads to them from the artery.

Where it is taught: Section C11, all three paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C11-BD2

Brain dump 2 · integration and application: A 33-year-old smoker with a black fingertip

A competent response contains

· Diagnosis is thromboangiitis obliterans, or Buerger disease.

· Heavy smoking at age 33 fits the near exclusive association with tobacco and the onset before about 35 to 40.

· Instep claudication fits distal disease of the tibial arteries and their branches rather than proximal large artery stenosis.

· A digital ulcer fits progressive occlusion of digital branches.

· Absent radial and posterior tibial pulses with normal brachial and femoral pulses fit segmental distal arterial occlusion.

· Cold-triggered colour change is the secondary Raynaud phenomenon that accompanies this disease.

· Migratory nodular phlebitis fits spread of the inflammation to adjacent veins.

· Upper limb involvement argues against atherosclerotic disease, which affects the large and medium arteries of the legs; atherosclerosis in a man of 33 would also be unusual.

· Further separating features: age under about 40 against over 50; segmental inflammatory thrombosis with microabscesses against lipid plaque with a fibrous cap; migratory superficial phlebitis and severe neural pain against calf or buttock claudication; risk factors in atherosclerosis extend beyond smoking.

· The vein nodules and the colour change show the process involves adjacent veins and the small digital arteries, not the arteries alone.

· Whether he keeps his digits depends on complete cessation of all tobacco use, because the disease rarely progresses once tobacco exposure stops.

Distinctions that matter

· Distal and upper limb involvement in a young smoker is the pattern that separates this disease from atherosclerotic peripheral arterial disease.

· Rest pain here is partly neural, which is why it is more severe than the ulcer size suggests.

Common misconceptions

· Calling this premature atherosclerosis because the patient smokes. The lesion, the vessels, and the vein and nerve involvement differ.

· Expecting a vascular procedure to settle the outcome. Continued smoking drives progression.

Where it is taught: Section C11, all three paragraphs and the comparison table, with Raynaud phenomenon in C10 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C12-BD1

Brain dump 1 · foundational retrieval: Sort the vasculitides by vessel size, then build giant cell arteritis

A competent response contains

· Vasculitis is inflammation of the vessel wall, and the Chapel Hill classification groups the vasculitides by the size of vessel mainly affected, which predicts the clinical picture.

· Large vessel disease narrows the aorta and its branches and causes limb or organ ischemia; medium vessel disease causes infarcts and aneurysms in organs; small vessel disease involves capillaries and venules, producing palpable purpura, glomerulonephritis, and alveolar hemorrhage.

· Large: giant cell arteritis, T cell granulomatous, age over 50 with polymyalgia rheumatica; Takayasu arteritis, T cell granulomatous, women under 40 to 50 and Asian.

· Medium: polyarteritis nodosa, immune complex in some cases, hepatitis B and ANCA negative; Kawasaki disease, T cell and antibody response to an unknown trigger, children under 5 with coronary aneurysm.

· Small ANCA: granulomatosis with polyangiitis, PR3-ANCA with granulomas, sinus, lung, kidney; microscopic polyangiitis, MPO-ANCA without granulomas, lung and kidney; eosinophilic granulomatosis with polyangiitis, eosinophils with MPO-ANCA in a minority, asthma and eosinophilia.

· Small immune complex: IgA vasculitis, IgA1 immune complexes, children after an upper respiratory infection.

· Variable: Behcet syndrome, neutrophilic inflammation, oral and genital ulcers with uveitis and HLA-B51.

· Giant cell arteritis almost never begins before age 50, affects women more often, and is most frequent in people of Northern European ancestry; it is T cell mediated granulomatous inflammation of large and medium arteries.

· Arteries involved: branches of the external carotid including the superficial temporal artery, the ophthalmic and posterior ciliary arteries, the vertebral arteries, and the aorta.

· Activated T cells and macrophages release cytokines including IL-6, which drives the acute-phase response.

· Symptoms: new headache with a tender thickened temporal artery from arteritis of that vessel; jaw claudication from masseter ischemia on chewing; visual loss from occlusion of the ophthalmic or posterior ciliary arteries causing anterior ischemic optic neuropathy, which can be sudden and permanent.

· The sedimentation rate is markedly raised, often above 50 mm/h, with high CRP.

· Biopsy shows granulomatous inflammation of the inner media with multinucleated giant cells and fragmentation of the internal elastic lamina; lesions are segmental skip lesions, so a short or unlucky biopsy can be negative.

· Polymyalgia rheumatica causes pain and morning stiffness of the shoulder and hip girdles and neck with a high sedimentation rate but normal creatine kinase and normal strength, because the target is periarticular synovium and bursae rather than muscle; about half of patients with giant cell arteritis have its symptoms, and a smaller fraction of patients with it develop giant cell arteritis.

· Aortic involvement in giant cell arteritis can later cause thoracic aneurysm.

Distinctions that matter

· Vessel size predicts the organ picture, and the immune mechanism predicts the laboratory findings, so both columns are needed.

· Polymyalgia rheumatica is inflammatory but not myopathic, which is why strength and creatine kinase are normal.

Common misconceptions

· Treating a negative temporal artery biopsy as excluding the disease. Skip lesions make false negatives common.

· Calling polymyalgia rheumatica a muscle disease because of the name. The inflamed tissue is periarticular.

Where it is taught: Section C12, all paragraphs and the vasculitis by vessel size table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C12-BD2

Brain dump 2 · integration and application: Jaw pain, a headache, and sudden loss of vision

A competent response contains

· Headache with a tender thickened temporal cord: granulomatous arteritis of the superficial temporal artery, a branch of the external carotid, thickens the wall and makes it tender and palpable.

· Jaw pain after a few minutes of chewing: jaw claudication from ischemia of the masseter, because the narrowed arteries cannot raise flow to meet the muscle's demand.

· Shoulder and hip aching with morning stiffness: polymyalgia rheumatica, inflammation of periarticular synovium and bursae.

· Sudden painless vision loss: occlusion of the ophthalmic or posterior ciliary arteries producing anterior ischemic optic neuropathy, which can be permanent.

· A sedimentation rate of 86 mm/h with high CRP: IL-6 released by activated T cells and macrophages drives the acute-phase response.

· Her girdle symptoms are not a muscle disease because strength and creatine kinase are normal; the inflamed tissue is periarticular synovium and bursae.

· A normal biopsy does not exclude the disease, because the lesions are segmental skip lesions and a short or unlucky sample can miss them; it also does not confirm an alternative diagnosis.

· Giant cell arteritis can involve the aorta, and that involvement can later cause a thoracic aneurysm.

Distinctions that matter

· Jaw claudication and headache are both ischemic, but one follows exertion of a muscle and the other follows inflammation of the vessel itself.

· A high sedimentation rate reports the acute-phase response rather than the site of disease, so it supports but does not localize.

Common misconceptions

· Expecting painful vision loss. The ischemic optic neuropathy of this disease is painless.

· Assuming a normal biopsy means the headache has another cause.

Where it is taught: Section C12, giant cell arteritis and polymyalgia rheumatica paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-BD1

Brain dump 1 · foundational retrieval: Takayasu arteritis from arch to pulse

A competent response contains

· Takayasu arteritis is a granulomatous vasculitis of large arteries, principally the aortic arch and its branches, and sometimes the descending and abdominal aorta, the renal arteries, and the pulmonary arteries.

· It most often affects women under about 40 to 50 and is more common in Asia.

· Histology resembles giant cell arteritis: mononuclear infiltrates, granulomas with giant cells, and destruction of the media; age is what mainly separates the two, so a granulomatous large-vessel arteritis in a young woman is Takayasu and in a patient over 50 it is giant cell arteritis.

· Chronic inflammation is followed by fibrosis of the media and adventitia and thickening of the intima, giving an irregularly thickened aortic wall with stenosis or occlusion of the origins of the great vessels.

· The disease often begins with fever, fatigue, weight loss, and arthralgia, with a raised sedimentation rate, before ischemic signs appear.

· Narrowing of the subclavian arteries at their origins makes pulses in one or both arms weak or absent, which gave the older name pulseless disease; blood pressure in the affected arm is lower so the two arms differ, arm claudication occurs with use, and bruits are heard over the subclavian and carotid arteries.

· Carotid and vertebral narrowing causes dizziness, syncope, visual disturbances, and stroke.

· Renal artery stenosis activates the renin-angiotensin system and causes hypertension, which can be masked when the pressure is measured in a narrowed arm.

· Aortic root involvement can dilate the root and cause aortic regurgitation, and pulmonary artery involvement can cause pulmonary hypertension.

Distinctions that matter

· Both large-vessel granulomatous arteritides look the same under the microscope, so age and the vessels involved carry the diagnosis.

· The inflammatory phase and the ischemic phase are separated in time, which is why constitutional symptoms come first.

Common misconceptions

· Trusting a normal blood pressure measured in a narrowed arm. The true central pressure can be much higher.

· Expecting the temporal artery to be involved. This disease centres on the arch and its branches.

Where it is taught: Section C13, all three paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-BD2

Brain dump 2 · integration and application: Two women with a weak arm pulse

A competent response contains

· The woman has Takayasu arteritis: granulomatous inflammation followed by intimal thickening and fibrosis narrows the origin of the left subclavian artery, so pressure distal to the stenosis is lower in that arm, with a bruit over the narrowed segment.

· The man has an aortic dissection: a false lumen in the media compresses the left subclavian artery, so flow and pressure fall in that arm.

· History separates them: months of fever, fatigue, weight loss, and a high sedimentation rate against sudden severe tearing pain radiating between the scapulae.

· Time course is the most useful discriminator because a chronic inflammatory stenosis develops over months while a dissection produces its findings within minutes.

· A pressure measured in her narrowed left arm underestimates central pressure, so hypertension can be masked; if renal artery stenosis is also present, significant hypertension could be missed entirely.

· Renal artery involvement activates the renin-angiotensin system and raises blood pressure.

· In a 68-year-old a granulomatous large-vessel arteritis would be giant cell arteritis, and age over 50 is the feature that decides it.

Distinctions that matter

· Unequal arm pressures is a shared finding with very different mechanisms: a fixed inflammatory stenosis against acute compression by a false lumen.

· Measuring blood pressure in both arms matters in both diseases, for opposite reasons.

Common misconceptions

· Treating unequal arm pressures as specific for dissection.

· Assuming a young woman cannot have severe hypertension because the cuff reading is low. The reading may be taken beyond a stenosis.

Where it is taught: Section C13, second and third paragraphs, with the dissection findings in C05 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-CP02

Comparison table: The two granulomatous large-vessel arteritides

A competent response contains

· Age: giant cell arteritis over 50 and almost never before; Takayasu usually under 40 to 50.

· Typical patient: older woman of Northern European ancestry; young woman, often Asian.

· Main vessels: carotid branches including the temporal and the ophthalmic and posterior ciliary arteries, the vertebral arteries, and the aorta; the aortic arch and its branches, with the descending and abdominal aorta, renal arteries, and pulmonary arteries at times.

· Key findings: new headache with a tender thickened temporal artery, jaw claudication, visual loss, and polymyalgia rheumatica; weak or absent arm pulses, unequal arm pressures, arm claudication, bruits over the subclavian and carotid arteries, and renin-mediated hypertension with renal involvement.

· Histology: both granulomatous with multinucleated giant cells; giant cell arteritis also shows fragmentation of the internal elastic lamina with segmental skip lesions, and Takayasu progresses to fibrosis of the media and adventitia with intimal thickening.

· Laboratory: markedly raised sedimentation rate, often above 50 mm/h, with high CRP in giant cell arteritis; raised sedimentation rate with constitutional symptoms in Takayasu.

· Late aortic complication: thoracic aneurysm in giant cell arteritis; aortic root dilation with regurgitation in Takayasu, and pulmonary hypertension with pulmonary artery involvement.

Distinctions that matter

· The histology column is nearly identical, which is exactly why the age and vessel rows carry the diagnosis.

· Both can involve the aorta, and the consequence differs: aneurysm in one and root dilation with regurgitation in the other.

Common misconceptions

· Expecting one of the two to be non-granulomatous.

· Filling the vessel row for giant cell arteritis with the temporal artery alone. It also involves ophthalmic, posterior ciliary, and vertebral arteries and the aorta.

Where it is taught: Section C13, comparison table, with the giant cell arteritis paragraphs in C12 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C14-BD1

Brain dump 1 · foundational retrieval: Polyarteritis nodosa, vessel by vessel

A competent response contains

· Polyarteritis nodosa is a systemic necrotizing vasculitis of small and medium muscular arteries, typically renal and visceral arteries, that spares the pulmonary arteries and does not involve arterioles, capillaries, or venules.

· Because the smallest vessels are spared it does not cause glomerulonephritis.

· It most often affects young to middle-aged adults.

· A substantial minority, historically about a third, have chronic hepatitis B, and in them immune complexes containing hepatitis B surface antigen deposit in the vessel wall, a type III mechanism.

· ANCA is characteristically negative.

· The lesion is segmental transmural inflammation with fibrinoid necrosis of the wall, initially rich in neutrophils, eosinophils, and mononuclear cells.

· Necrosis weakens the wall, producing small aneurysms that give involved arteries a beaded or nodular appearance, which is the source of the name, and it promotes thrombosis and infarction.

· The hallmark is that acute and healed fibrotic lesions coexist in the same patient and even the same artery, showing that injury occurs in waves.

· Angiography of the renal or mesenteric arteries shows multiple microaneurysms and stenoses.

· Manifestations: renal artery involvement causes renal infarcts and renin-mediated hypertension; mesenteric arteritis causes abdominal pain worse after meals, bleeding, and bowel infarction or perforation; inflammation of the vasa nervorum causes mononeuritis multiplex such as foot drop; skin shows painful nodules, livedo reticularis, and ulcers; testicular pain can occur; fever, malaise, and weight loss are common.

· Renal involvement with hypertension is the leading cause of death.

Distinctions that matter

· Medium-vessel disease produces infarcts and aneurysms, while small-vessel disease produces capillary inflammation with purpura and glomerulonephritis.

· The absence of lung involvement and of glomerulonephritis is positive information, because it tells you which vessels are spared.

Common misconceptions

· Expecting glomerulonephritis because the kidney is involved. The disease injures renal arteries and spares the glomerular capillaries.

· Looking for ANCA to confirm the diagnosis. It is characteristically negative.

Where it is taught: Section C14, all three paragraphs and the features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C14-BD2

Brain dump 2 · integration and application: Explain a patient with fever, foot drop, and bloody stools

A competent response contains

· Diagnosis is polyarteritis nodosa associated with chronic hepatitis B.

· Hypertension and the raised creatinine: arteritis of the renal and intrarenal arteries reduces glomerular perfusion and activates renin, and renal infarcts reduce function.

· Postprandial abdominal pain with dark stools: mesenteric arteritis causes ischemia that worsens when digestive demand rises, with mucosal bleeding.

· Foot drop: mononeuritis multiplex from infarction of nerves through involvement of the vasa nervorum.

· Painful calf nodules: inflamed, aneurysmal small and medium cutaneous arteries.

· Fever, malaise, and weight loss: systemic inflammation.

· No red cell casts and no hematuria because arterioles, capillaries, and venules are spared, so there is no glomerulonephritis; this confirms medium-vessel rather than small-vessel disease.

· Hepatitis B matters because immune complexes containing hepatitis B surface antigen deposit in the vessel wall as a type III mechanism, and those deposits would be found in the wall.

· Angiography would show multiple microaneurysms with stenoses, because fibrinoid necrosis weakens the wall at multiple points.

· Hemoptysis with red cell casts and a positive ANCA would point to an ANCA-associated small-vessel vasculitis such as microscopic polyangiitis.

· Acute and healed lesions in the same artery show that the injury occurs in waves rather than all at once.

Distinctions that matter

· A rising creatinine without casts or hematuria is renal artery disease, while casts and hematuria mean glomerular capillary disease.

· Hepatitis B supports an immune complex mechanism, which is deposit-rich, unlike the pauci-immune ANCA diseases.

Common misconceptions

· Interpreting any renal failure in a vasculitis as glomerulonephritis.

· Expecting lung findings. The pulmonary arteries are spared, so normal chest imaging fits the diagnosis.

Where it is taught: Section C14, all three paragraphs and the features table, with the ANCA diseases in C16 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C15-BD1

Brain dump 1 · foundational retrieval: Kawasaki disease and the coronary arteries

A competent response contains

· Kawasaki disease, or mucocutaneous lymph node syndrome, is an acute, self-limited vasculitis of medium arteries that mainly affects children under 5 and is most common in children of East Asian, especially Japanese, ancestry.

· It is the leading cause of acquired heart disease in children in developed countries.

· The trigger is unknown; the leading view is that an infectious agent provokes an exaggerated T cell and antibody response in genetically susceptible children, with activated macrophages and cytokines damaging the vessel wall.

· The arteritis is transmural with mixed inflammatory cells, and fibrinoid necrosis is usually less prominent than in polyarteritis nodosa.

· Diagnosis is clinical: fever lasting at least 5 days plus most of the five other principal features, with fever plus at least four of the five supporting classic disease.

· Features: high fever for 5 days or more; bilateral nonexudative conjunctival injection sparing the limbus; red cracked lips with strawberry tongue and diffuse oral redness; polymorphous rash often on trunk and perineum; red swollen hands and feet with later periungual desquamation; cervical lymphadenopathy, usually unilateral, at least 1.5 cm.

· Peeling of the fingers and toes typically follows in the second to third week, and irritability is marked.

· Laboratory: raised sedimentation rate and CRP, leukocytosis, and thrombocytosis after the first week.

· The key complication is coronary artery aneurysm in roughly a quarter of untreated children: inflammation destroys the internal elastic lamina and media of the coronary arteries, which then dilate.

· Inside an aneurysm flow is slow and turbulent and the high platelet count adds to the tendency to thrombose, so coronary thrombosis can cause myocardial infarction or sudden death in a young child; aneurysms can rupture and healing can leave stenosis.

· Myocarditis, pericarditis, and valvular regurgitation can occur in the acute phase.

Distinctions that matter

· The mucocutaneous findings are temporary while the coronary lesion is the lasting problem, so the visible illness and the dangerous one are not the same.

· Fibrinoid necrosis is prominent in polyarteritis nodosa and usually less so here, although both are medium-vessel diseases with aneurysms.

Common misconceptions

· Treating the rash and mucosal findings as the reason the disease matters. The coronary arteries decide the outcome.

· Expecting exudative conjunctivitis. The conjunctival injection is nonexudative and spares the limbus.

Where it is taught: Section C15, all three paragraphs and the principal features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C15-BD2

Brain dump 2 · integration and application: A red-eyed, red-tongued, miserable toddler

A competent response contains

· Diagnosis is Kawasaki disease.

· Criteria counted: fever for six days, which exceeds the 5-day requirement, plus conjunctival injection, lip and oral changes with strawberry tongue, polymorphous trunk and perineal rash, red swollen extremities, and cervical lymphadenopathy of 2 cm, which is all five other features, so the criteria are met.

· The eye finding is bilateral nonexudative conjunctival injection, distinguished by the absence of discharge and by sparing of the limbus.

· In the second to third week periungual desquamation is expected, with peeling of the fingers and toes.

· The platelet count is raised because thrombocytosis develops after the first week as part of the acute inflammatory response.

· The coronary arteries are the concern: inflammation destroys the internal elastic lamina and the media, so the artery dilates into an aneurysm.

· Inside the aneurysm flow is slow and turbulent and the high platelet count adds to the tendency to clot, so coronary thrombosis can cause myocardial infarction or sudden death in a young child.

· Scarlet fever is caused by group A streptococcal infection, usually with exudative pharyngitis, and although it also gives a strawberry tongue and rash it does not cause conjunctivitis or the swollen desquamating extremities, and it does not cause coronary aneurysms.

Distinctions that matter

· Strawberry tongue and rash are shared with scarlet fever, and conjunctival injection with extremity changes are not.

· Thrombocytosis is part of the course rather than an unrelated finding, and its timing after the first week is informative.

Common misconceptions

· Dismissing the diagnosis because a single feature is missing. Fever plus four of the five supports classic disease.

· Attributing the irritability to the fever alone. Marked irritability is itself part of the illness.

Where it is taught: Section C15, second and third paragraphs and the principal features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-BD1

Brain dump 1 · foundational retrieval: How ANCA injures a capillary, and the three diseases that follow

A competent response contains

· ANCA target enzymes stored in neutrophil granules: PR3-ANCA against proteinase 3 gives a cytoplasmic pattern called c-ANCA, and MPO-ANCA against myeloperoxidase gives a perinuclear pattern called p-ANCA.

· Mechanism: infection or cytokines prime the neutrophil; PR3 and MPO move to the cell surface; ANCA bind them there; the bound neutrophil degranulates and releases reactive oxygen species while adhering to the endothelium; small vessels, meaning arterioles, capillaries, and venules, are injured.

· Pauci-immune means little or no immunoglobulin or complement is deposited in the lesion, and these diseases are pauci-immune because injury depends on activated cells rather than on deposits.

· An immune complex vasculitis instead shows immunoglobulin and complement in the vessel wall on immunofluorescence.

· The kidney and lung are hit hardest because they are rich in capillaries: necrotizing crescentic glomerulonephritis causes hematuria, red cell casts, and rapidly progressive renal failure, and pulmonary capillaritis causes alveolar hemorrhage with hemoptysis.

· Combined lung hemorrhage and glomerulonephritis is a pulmonary-renal syndrome, which also occurs with anti-GBM disease, identified by linear IgG along the basement membrane on immunofluorescence.

· Granulomatosis with polyangiitis: PR3-ANCA in most; necrotizing granulomas; chronic sinusitis, nasal ulcers, septal perforation, saddle nose, and otitis; lung nodules that cavitate with hemorrhage; pauci-immune crescentic glomerulonephritis; the classic triad of upper airway, lung, and kidney.

· Microscopic polyangiitis: MPO-ANCA in most; no granulomas; upper airway usually spared; capillaritis with hemorrhage; pauci-immune crescentic glomerulonephritis; like the former without granulomas or upper airway disease.

· Eosinophilic granulomatosis with polyangiitis: MPO-ANCA in a minority, about 40%; eosinophil-rich granulomas; allergic rhinitis and nasal polyps; asthma with transient infiltrates; kidney involvement less common and milder; blood eosinophilia, mononeuritis multiplex, and cardiac involvement.

Distinctions that matter

· Granulomas, eosinophils, and the ANCA type separate the three diseases, and no single one of the three is sufficient by itself.

· Pauci-immune and immune complex disease can look similar clinically, and immunofluorescence settles which it is.

Common misconceptions

· Expecting immune deposits in a vasculitis with glomerulonephritis. These diseases are pauci-immune.

· Assuming ANCA must be positive in eosinophilic granulomatosis with polyangiitis. It is positive in fewer than half.

Where it is taught: Section C16, first and second paragraphs and the ANCA comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-BD2

Brain dump 2 · integration and application: Three patients who all cough blood

A competent response contains

· Patient 1: granulomatosis with polyangiitis, placed there by destructive upper airway disease with a saddle nose, cavitating lung nodules, glomerulonephritis, and PR3-ANCA, which is the classic triad of upper airway, lung, and kidney with necrotizing granulomas.

· Patient 2: microscopic polyangiitis, placed there by lung hemorrhage and glomerulonephritis with palpable purpura and MPO-ANCA but no granulomas and no upper airway destruction.

· Patient 3: eosinophilic granulomatosis with polyangiitis, placed there by long-standing asthma and nasal polyps progressing to marked eosinophilia, transient infiltrates, mononeuritis multiplex, and cardiac involvement.

· Each renal biopsy would show necrotizing crescentic glomerulonephritis with little or no immunoglobulin or complement, which is what pauci-immune means, and in the third patient renal involvement is less common and milder.

· Patient 1 has upper airway destruction because necrotizing granulomas form in the upper and lower respiratory tract; patient 2 has small-vessel injury without granulomas, so the airway is usually spared.

· In patient 3 blood and tissue eosinophilia drives injury through eosinophil granule proteins, ANCA is positive in fewer than half of such patients and MPO-ANCA in about 40%, and cardiac involvement is the main cause of death.

· Anti-GBM disease would be suspected in lung hemorrhage with glomerulonephritis when ANCA is negative, and immunofluorescence showing linear IgG along the glomerular basement membrane would identify it.

Distinctions that matter

· All three can injure lung and kidney, so granulomas, the upper airway, and eosinophils are what separate them.

· Pulmonary-renal syndrome names a pattern rather than a disease, and immunofluorescence distinguishes pauci-immune from linear IgG disease.

Common misconceptions

· Reading a saddle nose as trauma rather than as destruction of cartilage by necrotizing granulomas.

· Assuming asthma is incidental in patient 3. It is the first phase of the disease.

Where it is taught: Section C16, second and third paragraphs and the ANCA comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-Q04

Board-style question: Sinuses, cavities, and casts

Correct answer: B. PR3-ANCA with a cytoplasmic immunofluorescence pattern

Why it is right: Upper airway destruction with saddle nose, cavitating lung nodules, and pauci-immune necrotizing crescentic glomerulonephritis is granulomatosis with polyangiitis, in which PR3-ANCA giving a cytoplasmic immunofluorescence pattern is present in most patients.

A is wrong: Linear IgG along the glomerular basement membrane identifies anti-GBM disease, which causes a pulmonary-renal syndrome but not destructive sinus disease or cavitating nodules, and its biopsy is not pauci-immune.

C is wrong: IgA-containing complexes in the mesangium identify IgA vasculitis or IgA nephropathy, which give palpable purpura and mesangial disease rather than saddle nose and lung cavities, and the lesion is deposit-rich.

D is wrong: Hepatitis B surface antigen in circulating immune complexes is associated with polyarteritis nodosa, which spares the pulmonary arteries and the small vessels and therefore causes no glomerulonephritis.

A competent response contains

· Destructive upper airway disease with a saddle nose, cavitating lung nodules, and crescentic glomerulonephritis is the classic triad of granulomatosis with polyangiitis.

· Almost no immunoglobulin or complement on immunofluorescence makes the lesion pauci-immune, which fits ANCA disease, and PR3-ANCA with a cytoplasmic pattern is positive in most such patients.

Distinctions that matter

· Pauci-immune disease points to ANCA, while linear IgG and granular IgA deposits point to other mechanisms entirely.

Common misconceptions

· Expecting deposits to be found because there is glomerulonephritis.

Where it is taught: Section C16 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C17-BD1

Brain dump 1 · foundational retrieval: IgA vasculitis and Behcet syndrome from mechanism to sign

A competent response contains

· IgA vasculitis, formerly Henoch-Schonlein purpura, is the most common systemic vasculitis of childhood, typically between about 3 and 15 years, and often follows an upper respiratory infection.

· Patients have an excess of galactose-deficient IgA1, which is recognized by autoantibodies to form immune complexes.

· The complexes deposit in small vessels of the skin, gut, joints, and glomerular mesangium, where they activate complement through the alternative and lectin pathways; this is a type III immune complex mechanism, and immunofluorescence shows IgA in vessel walls and the mesangium.

· Skin: dermal leukocytoclastic vasculitis gives palpable purpura over the buttocks and the extensor surfaces of the legs, where gravity and hydrostatic pressure favour deposition.

· Joints: arthralgia or arthritis, usually of the knees and ankles.

· Gut: vasculitis causes colicky abdominal pain and bleeding and predisposes to intussusception, often ileoileal.

· Kidney: mesangial glomerulonephritis identical to IgA nephropathy, with hematuria and sometimes proteinuria, more often severe in adults.

· Platelet count and coagulation tests are normal because the purpura comes from inflamed vessels, which excludes thrombocytopenic purpura.

· Behcet syndrome is a variable-vessel vasculitis affecting arteries and veins of any size, most common along the old Silk Road from the Mediterranean and Middle East to Japan, associated with HLA-B51, and driven by neutrophil hyperactivity.

· Its features: recurrent painful oral aphthous ulcers in nearly all patients, genital ulcers, uveitis that can cause blindness and sometimes shows a hypopyon, erythema nodosum and pathergy in the skin, venous thrombosis and arterial aneurysms, and neurologic involvement.

· Pathergy is an exaggerated pustular response to a needle prick.

· Palpable purpura with normal platelets in a child points to IgA vasculitis; recurrent mucosal ulcers with uveitis point to Behcet syndrome.

Distinctions that matter

· One disease is deposit-driven and confined to small vessels, the other is neutrophil-driven and affects vessels of any size.

· Purpura from vessel inflammation and purpura from low platelets look similar, and the platelet count separates them.

Common misconceptions

· Expecting a low platelet count because the child has purpura.

· Treating the renal lesion as unrelated to IgA nephropathy. It is identical.

Where it is taught: Section C17, all three paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C17-BD2

Brain dump 2 · integration and application: Purpura in a child and ulcers in a young adult

A competent response contains

· The boy has IgA vasculitis following an upper respiratory infection.

· Palpable purpura on buttocks and the backs of the legs: dermal leukocytoclastic vasculitis where galactose-deficient IgA1 immune complexes deposit, favoured at those sites by gravity and hydrostatic pressure.

· Swollen painful knees and ankles: immune complex deposition in the joints giving arthralgia or arthritis.

· Colicky abdominal pain: gut vasculitis with ischemia of the bowel wall.

· Dark stools: bleeding from the inflamed gut vessels.

· Blood in the urine: mesangial glomerulonephritis from IgA deposition in the glomerular mesangium.

· A normal platelet count matters because it shows the purpura comes from inflamed vessels rather than from low platelets, which rules out thrombocytopenic purpura.

· He is at risk of intussusception, often ileoileal.

· Kidney biopsy would show mesangial IgA deposition with a mesangial glomerulonephritis identical to IgA nephropathy.

· The second patient has Behcet syndrome, associated with HLA-B51, driven by neutrophil hyperactivity; the needle finding is pathergy and the layer of pus in the anterior chamber is a hypopyon.

· Behcet syndrome is called a variable-vessel vasculitis because it affects arteries and veins of any size, which is why he has both a venous thrombosis and a risk of arterial aneurysm, whereas IgA vasculitis is confined to small vessels.

Distinctions that matter

· Both diseases give skin lesions, and purpura on dependent surfaces differs from erythema nodosum and pathergy.

· Venous thrombosis and arterial aneurysm in the same patient is what variable-vessel involvement looks like in practice.

Common misconceptions

· Ordering a platelet count to confirm the diagnosis in the boy. It is normal, and its value is in excluding another cause.

· Reading the pustule at the needle site as an infection rather than as pathergy.

Where it is taught: Section C17, first, second, and third paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.